KRITTITEE NAULKHAO

Bangkok, Thailand | (+66) 94-291-4565 | krittitee.n07@gmail.com
t6523.github.io | linkedin.com/in/krittiteenaulkhao

EDUCATION

Chulalongkorn University Bangkok, Thailand
B.E. Computer Engineering (Information & Communication) *Expected October 2027*

- GPA: 4.00/4.00, Full Academic Scholarship.

HONORS & AWARDS

- International Mathematical Olympiad (IMO) – Silver Medal, 2024
- Asia Pacific Mathematics Olympiad (APMO) – Bronze Medal, 2023

RELEVANT COURSEWORK

Data Science, Artificial Intelligence, Probability & Statistics for Data Analysis, Linear Algebra, Data Structures & Algorithms, Discrete Structures, Multivariable Calculus, Natural Language Processing (in progress).

TECHNICAL SKILLS

- **Languages:** Python, Java, SQL
- **ML & Data:** PyTorch, Scikit-Learn, XGBoost, LightGBM, CatBoost, Optuna, Albumentations, Pandas
- **Concepts:** Machine Learning, Deep Learning, NLP, Reinforcement Learning, Statistics, A/B Testing, Regression
- **Tools:** Git, Docker, DuckDB, ONNX, KenLM

PROJECTS

Double DQN Minesweeper Agent dqn-minesweeper-solver
PyTorch, Deep Reinforcement Learning, CNN, ONNX

- Built a Double DQN from scratch: 10.36 turns to clear a board vs. 22.9 for random play, 9.9 for an exact solver.
- Ran 3-seed ablations and deployed the model in the browser through ONNX.

A/B Test Planning and Analysis Toolkit ab-test-toolkit
Python, NumPy, SciPy, Statistics, Monte Carlo simulation

- Built a statistics toolkit for A/B tests: power analysis, hypothesis testing, confidence intervals, launch verdicts.
- Monte Carlo simulation verified 5.1% false positives and 95.0% CI coverage; peeking raised them to 19.2%.

Bangkok Airbnb Price Prediction bangkok-airbnb-pricing
Python, Regression, DuckDB (SQL), LightGBM, XGBoost, CatBoost, Optuna, SHAP, Makefile, Docker

- Built a regression pipeline (DuckDB SQL features, tuned boosted ensemble): 17% lower error than linear.
- Used grouped cross-validation, bootstrap CIs, and SHAP to explain price drivers (bedrooms, guests, stay rules).
- Made the pipeline reproducible with a Makefile and Docker.

Predictive Keyboard (NLP, Next-Word Prediction) predictive_keyboard
NLP, PyTorch, LoRA fine-tuning, KenLM n-gram language model, ensemble blending

- Raised accuracy from 60.1% (n-gram) to 78.6% by blending a fine-tuned LLM, a zero-shot LLM, and KenLM.
- Handled symbols, numbers, and words as separate tracks: majority vote, number n-gram, and the blend.

EXPERIENCE

I-Square Hackathon (Google & Gulf Sponsored) 2025
Machine Learning Mentor *Bangkok, Thailand*
PyTorch, Scikit-Learn, YOLO, Albumentations

- Built baseline deep learning models: ResNet50 classification and a YOLO / Faster R-CNN segmentation pipeline.
- Reduced class imbalance by oversampling minority classes with Albumentations augmentation.
- Mentored 20+ teams on machine learning pipelines: architecture, hyperparameter tuning, IoU optimization.

COURSES & CERTIFICATIONS

- CS50's Introduction to Artificial Intelligence with Python (Harvard CS50 AI)
- Introduction to Data Science (HarvardX CS109x)
- Introduction to Probability (HarvardX STAT110x)
- SQL for Data Science (IBM)